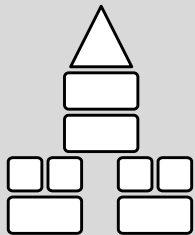
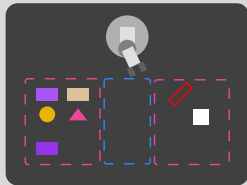


Input



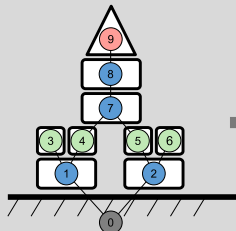
Task Specification

Pickup Area
Assembly Area

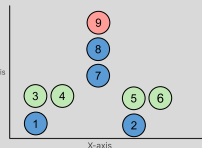


Initial Scene

Planning



1. Graph Generation



2.1 Branch Clustering



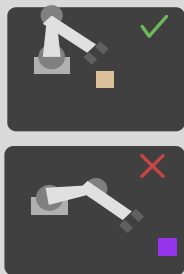
Executing Timeline

2.2 Layer-Based Cutting

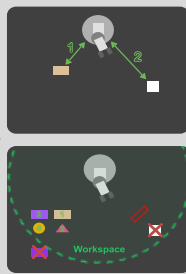
Executing Timeline



2. Two-Level Clustering 3. Execution Sequence



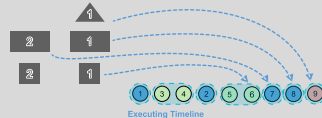
0.1 Manipulability-Aware Ordering



0.2 Manipulability-Ranked Candidates

0. Scene Grounding

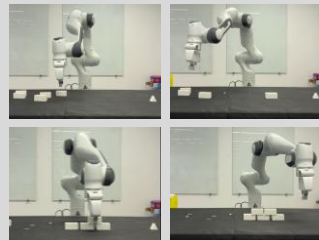
Solving & Executing



Execution-Order Embedding



LGP Solving with Active Constraints



Real Robot Execution